Branch and Bound

Job Sequencing using Branch and Bound:

Given n jobs with profit, execution time, and deadline, achieve the schedule which maximizes the profit.

- You are given a set of jobs.
- Each job has a defined deadline, execution time and some profit associated with it.
- The profit of a job is given only when that job is completed within its deadline.

Select optimal subset J with an optimal penalty for the following data. What will be the penalty corresponding to the optimal solution?

Job	Penalty	Deadline	Time
1	5	1	1
2	10	3	2
3	6	2	1
4	3	1	1

Solution:

Bounding function

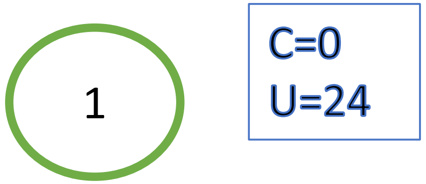
Upper bound:

u=sum of all penalties except that included in solution.

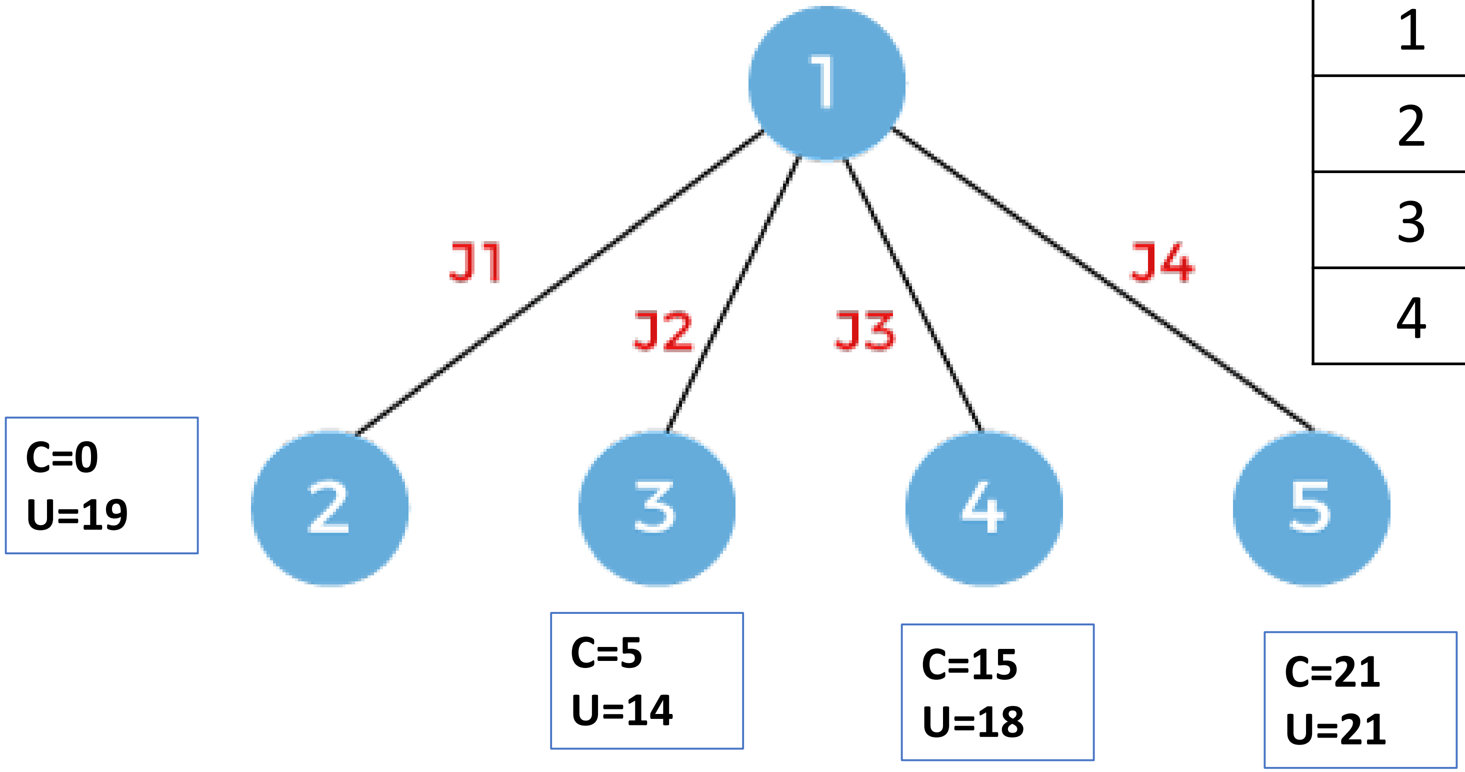
Cost function:

c = sum of penalties till the last job considered.

GUB= 24



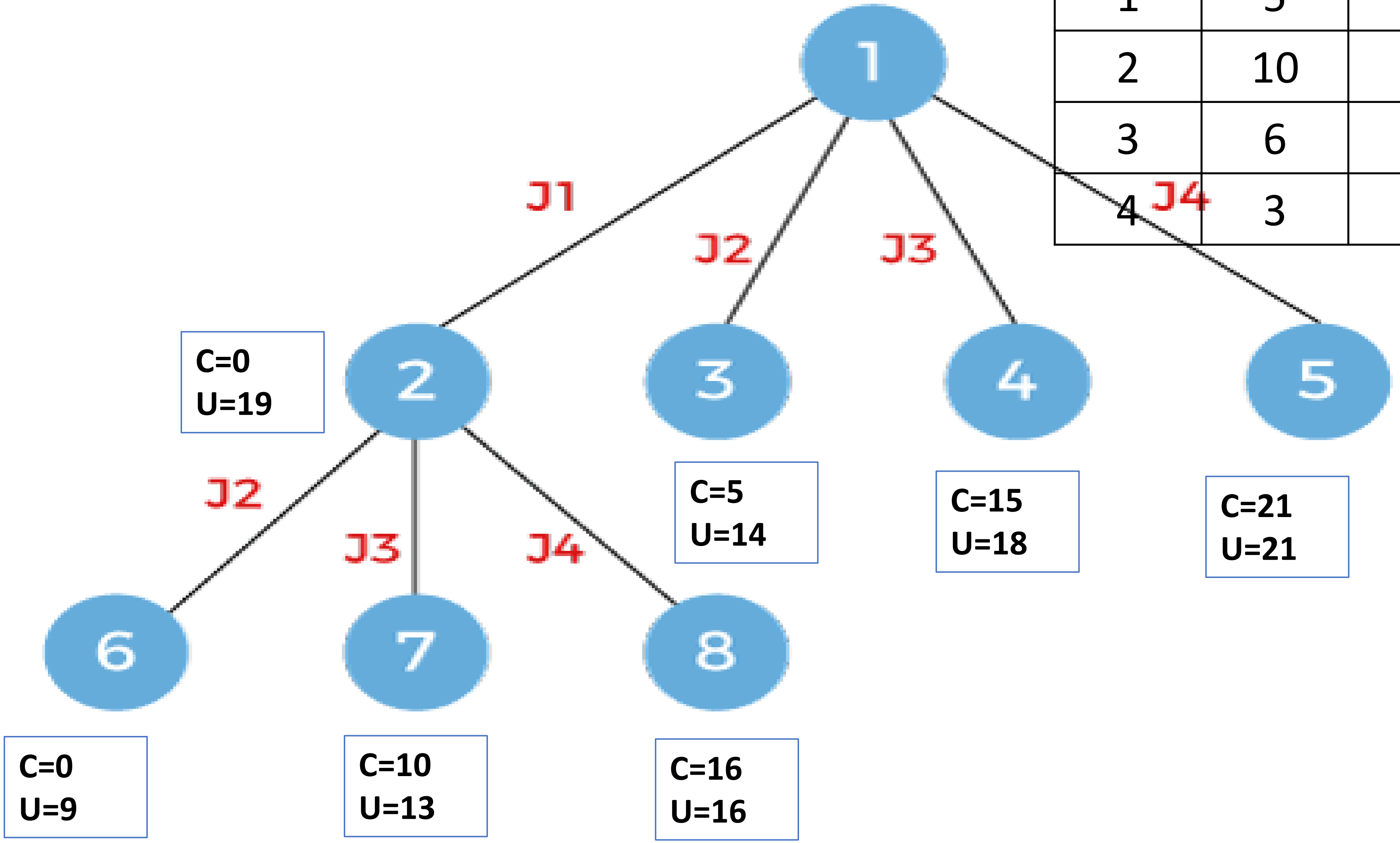
- GUB=14



Job	Penalty	Deadline	Time
1	5	1	1
2	10	3	2
3	6	2	1
4	3	1	1

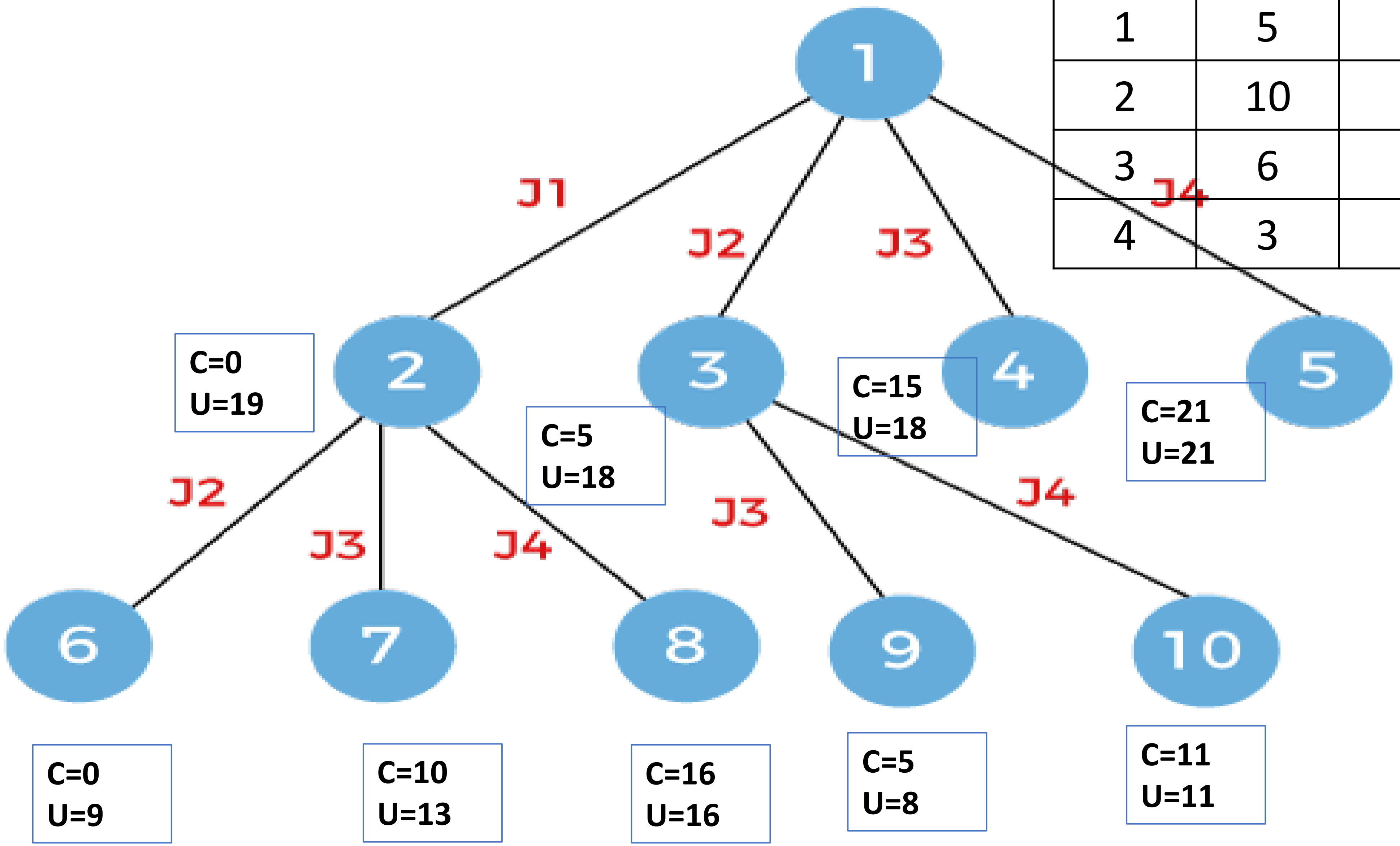
- **GUB=9**

Job	Penalty	Deadline	Time
1	5	1	1
2	10	3	2
3	6	2	1
4	3	1	1

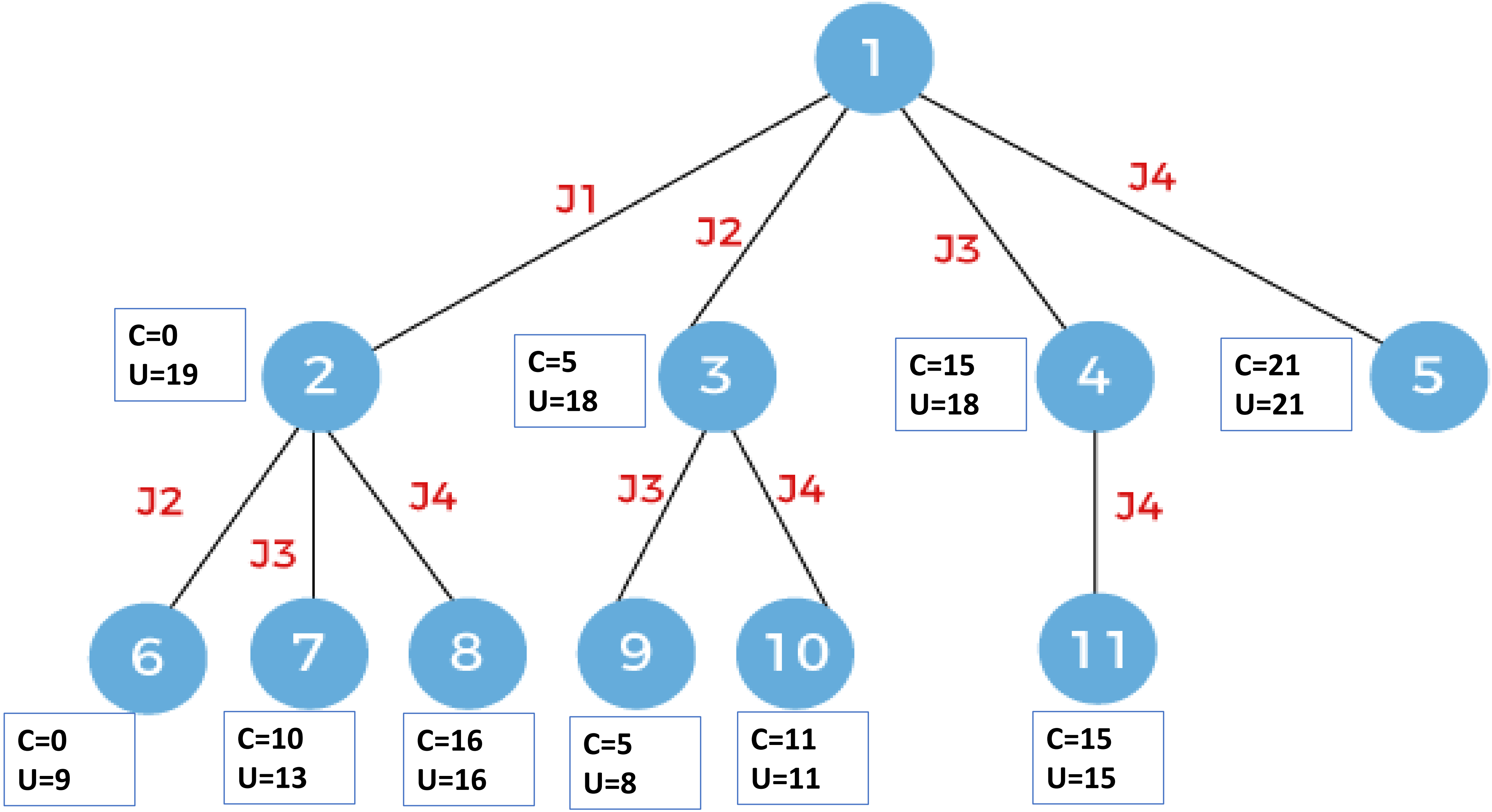


GUB=8

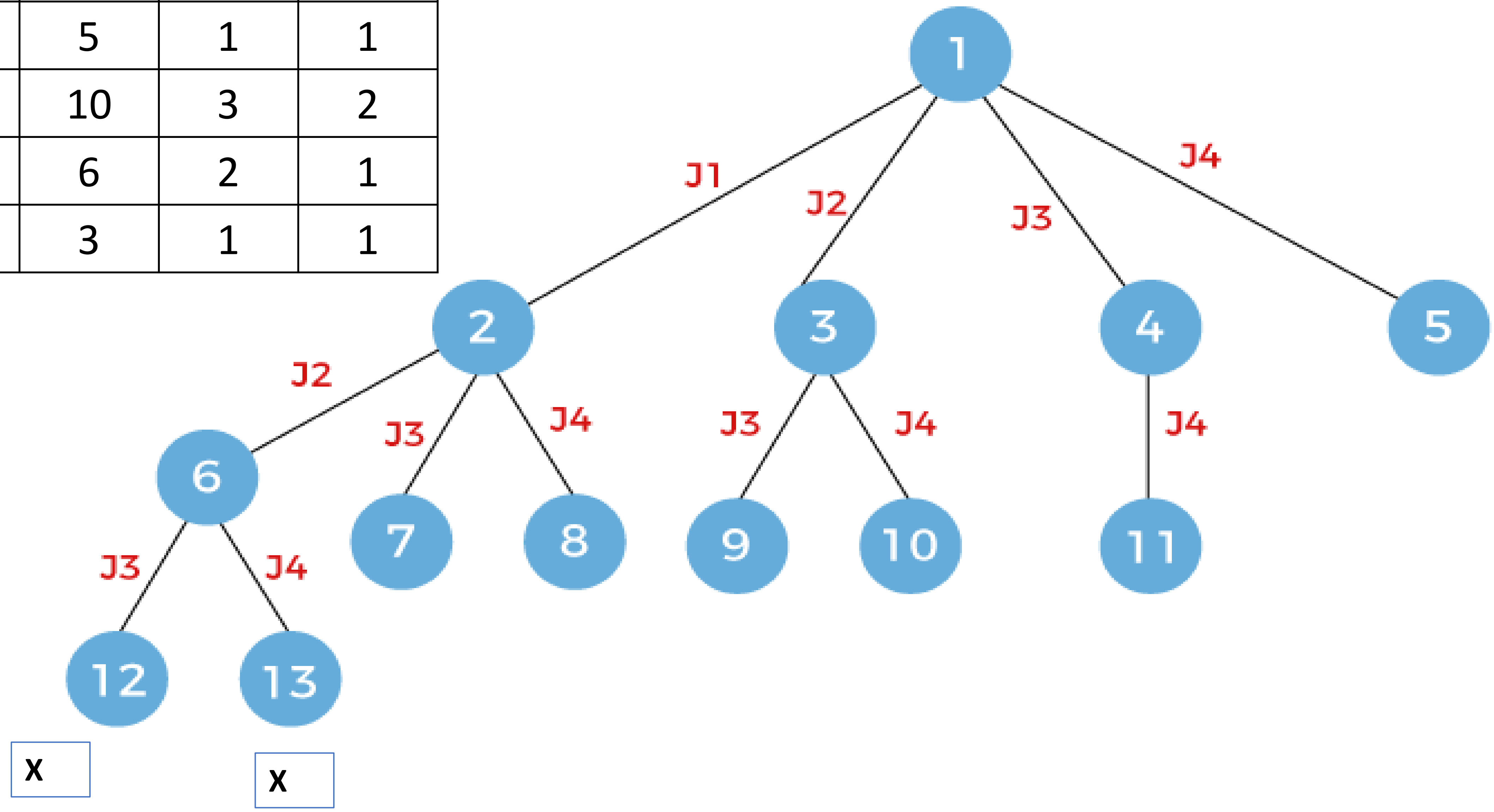
Job	Penalty	Deadline	Time
1	5	1	1
2	10	3	2
3	6	2	1
4	3	1	1



GUB=8


1
J1
J2
J3
J4
C=0
U=19
2
C=5
U=18
3
C=15
U=18
4
C=21
U=21
5
J2
J3
J4
J3
J4
J4
6
7
8
9
10
11
C=0
U=9
C=10
U=13
C=16
U=16
C=5
U=8
C=11
U=11
C=15
U=15

Job	Penalty	Deadline	Time
1	5	1	1
2	10	3	2
3	6	2	1
4	3	1	1



Job	Penalty	Deadline	Time
1	5	1	1
2	10	3	2
3	6	2	1
4	3	1	1